Exercise 2.23. Suppose that p and q are distinct primes, and a is a positive integer. Prove that if $p \mid a$ and $q \mid a$, then $pq \mid a$.

Solution.

Let p and q be distinct primes, and suppose

$$p \mid a \qquad \text{and} \qquad q \mid a.$$

We want to prove that

$$pq \mid a.$$

Since $p \mid a$, there exists an integer k such that

$$a = pk.$$

Because $q \mid a$, we also have

$$q \mid pk.$$

Since p and q are distinct primes,

$$\gcd(p, q) = 1.$$

By Euclid's lemma, if $q \mid pk$ and $\gcd(q, p) = 1$, then

$$q \mid k.$$

Therefore, for some integer m,

$$k = qm.$$

Substituting this into $a = pk$, we get

$$a = p(qm) = pqm.$$

Hence,

$$\boxed{pq \mid a}.$$

Therefore, if distinct primes p and q both divide a, then their product pq also divides a.